

Data Collection & Methodology

Mbala Municipal Council Web Dataset
CSC 4792 Data Mining and Warehousing — Mini Project
Project Team 17 — Mbala Municipal Council

1. Purpose of the Data Collection

The purpose of this project was to collect and curate publicly available information associated with Mbala Municipal Council for the CSC 4792 Data Mining and Warehousing mini-project. The resulting dataset is intended to support data mining, exploratory analysis, text analysis, information retrieval and further study of municipal information.

The project specification identifies municipal information such as CDF allocations and projects, budgets and revenue information, Integrated Development Plans, ward and administrative information, and council reports or resolutions as relevant categories.

2. Data Source

The primary source was the official Mbala Municipal Council website. The project used the supplied scraped page-level extract rather than claiming to represent every page or document available on the live website. The source records retained the original webpage URL, allowing each record to be traced back to its originating council page.

Item	Specification
Council	Mbala Municipal Council
Project	CSC 4792 Data Mining and Warehousing Mini Project
Project Team	Project Team 17
Primary source	Mbala Municipal Council official website
Source extract	42 scraped page-level records
Main dataset	42 records × 8 columns
File format	CSV with pipe () delimiter

3. Data Collection Approach

The collection process began with the supplied CSV extract of Mbala Municipal Council webpages. Each source record represented a webpage or post and included metadata such as URL, page/post identifiers, title, date, topic labels and extracted text. The original URL was preserved as a provenance field.

The collection and curation workflow followed these stages:

- Identify the assigned council and establish the official council website as the source.
- Use the supplied page-level CSV extract as the working source dataset.
- Retain webpage URLs and source identifiers to preserve provenance.
- Review available titles, topics and extracted text to identify relevant municipal information.
- Curate explicit CDF/LGEF project, asset, empowerment and bursary information where such information was present in the supplied records.
- Do not estimate, invent or infer unavailable figures or facts.

4. Data Fields Collected

Field	Collection / interpretation
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url	Original webpage URL; used for source provenance.
page_id	Identifier associated with the webpage record, where available.
post_id	Identifier associated with the original post/content record, where available.
title	Title of the webpage, post or publication.
date	Date associated with the webpage or publication, where available.
topics	Topic or category labels associated with the webpage record.
text_content	Textual content extracted from the corresponding webpage.
text_length_chars	Number of characters in the cleaned text_content field; derived during processing.

5. Data Processing and Preparation

The data preparation workflow was implemented in Python using pandas and documented in a Jupyter Notebook. The workflow included loading the pipe-delimited source file, inspecting its structure, standardising identifiers, converting dates, normalising whitespace, recalculating text length, profiling topic labels and performing validation checks.

No.	Stage	Method
1	Load	Load the source CSV using the required pipe () separator.
2	Inspect	Inspect dimensions, columns, data types and available values.
3	Standardise	Standardise identifier and date representations where applicable.
4	Normalise text	Handle missing text values and normalise repeated/leading/trailing whitespace.
5	Derive text length	Recalculate text_length_chars from the final text_content field.
6	Profile topics	Split and inspect topic labels for frequency and category analysis.
7	Validate	Check record counts, duplicate URLs, duplicate identifiers and missing values.
8	Export	Save the final cleaned dataset as a pipe-delimited CSV without the dataframe index.

6. CDF/LGEF Structured Curation

A structured CDF/LGEF dataset was curated from explicit development information contained in the supplied records. A structured record represents an explicit project, asset, empowerment or bursary item. Documented fields include record identifier, category, project/programme description, ward, CDF year/period, amount in Kwacha, reported progress, beneficiaries or units, sector and source URL.

Where a value was not supplied by the source, it was left unavailable rather than being calculated or guessed.

7. Quality Control and Validation

Quality control was performed after preparation to confirm that the resulting data remained consistent and usable. Checks included record counts, duplicate URL checks, duplicate structured record identifiers, missing-value inspection and checks of numeric fields in the structured development dataset.

The final page-level dataset contains 42 records and 8 variables: url, page_id, post_id, title, date, topics, text_content and text_length_chars.

8. File and Storage Conventions

The cleaned page-level dataset is stored as **db-unza26-csc4792-mbala_municipal_council_cleaned.csv**. CSV files use the pipe character (|) as the separator, and the exported cleaned file does not include the pandas dataframe index.

9. Provenance, Ethics and Limitations

The dataset is based on publicly available municipal web information and the supplied scraped extract. Source URLs are retained to support provenance and future verification. The dataset does not claim to be a complete archive of the council website. Some pages may refer to downloadable documents without containing their full contents; therefore complete budgets, full IDPs, complete WDC records or every CDF beneficiary/project should not be inferred from this dataset.

A key methodological principle was not to fabricate missing information. Blank or unavailable values represent information that was not provided in the supplied source extract.

10. Reproducibility

The workflow is reproducible through the project's Jupyter Notebook and supporting files. The project structure separates raw data from processed data, keeps the notebook in the notebooks directory, and stores supporting documentation in the documents directory. The cleaned CSV can be regenerated from the source data by following the documented notebook workflow.

11. Summary of the Methodology

The project used a supplied scrape of the official Mbala Municipal Council website, preserved source provenance, reviewed page metadata and extracted text, curated explicit development-related information, standardised and validated the data using Python/pandas, and exported the cleaned result as a pipe-delimited CSV. The methodology prioritises traceability, reproducibility and faithful representation of the source material.

References

1. CSC 4792 Data Mining and Warehousing Mini Project Practical Assignment, University of Zambia.
2. Mbala Municipal Council. Official website.
3. Project Team 17 Mbala Municipal Council dataset documentation and Jupyter Notebook.